Q1. JEE Main 2026 (23 January Shift 1)

The correct statements from the following are :

A. Ionic radii of trivalent cations of group 13 elements decreases down the group.

B. Electronegativity of group 13 elements decreases down the group.

C. Among the group 13 elements, Boron has highest first ionisation enthalpy.

D. The trichloride and triiodide of group 13 elements are covalent in nature.

Choose the correct answer from the options given below :

(1) C and D Only
(2) A and D Only
(3) A and C Only
(4) B and D Only

Q2. JEE Main 2026 (28 January Shift 2)

Consider the following reactions

$$Na_2B_4O_7 \xrightarrow{\Delta} 2X + Y$$

$$CuSO_4 + Y \xrightarrow{\text{Non-Luminous flame}} Z + SO_3$$

$$2Z + 2X + \text{Carbon} \xrightarrow{\text{Luminous flame}} 2Q + Na_2B_4O_7 + CO$$

The oxidation states of Cu in Z and Q, respectively are :

(1) +1 and +1
(2) +2 and +2
(3) +1 and +2
(4) +2 and +1

Q3. JEE Main 2026 (22 January Shift 2)

Given below are two statements :

Statement I: Elements ' X ' and ' Y ' are the most and least electronegative elements, respectively among N, As, Sb and P. The nature of the oxides X_2O_3 and Y_2O_3 is acidic and amphoteric, respectively.

Statement II : BCl_3 is covalent in nature and gets hydrolysed in water. It produces $[B(OH)_4]^-$ and $[B(H_2O)_6]^{3+}$ in aqueous medium.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are false
(2) Both Statement I and Statement II are true
(3) Statement I is true but Statement II is false
(4) Statement I is false but Statement II is true

Q4. JEE Main 2026 (23 January Shift 2)

It is noticed that Pb^{2+} is more stable than Pb^{4+} but Sn^{2+} is less stable than Sn^{4+}. Observe the following reactions.

$$PbO_2 + Pb \rightarrow 2PbO; \Delta_rG^\circ(1)$$

$$SnO_2 + Sn \rightarrow 2SnO; \Delta_rG^\circ(2)$$

Identify the correct set from the following

(1) $\Delta_rG^\circ(1) < 0; \Delta_rG^\circ(2) < 0$
(2) $\Delta_rG^\circ(1) > 0; \Delta_rG^\circ(2) > 0$
(3) $\Delta_rG^\circ(1) < 0; \Delta_rG^\circ(2) > 0$
(4) $\Delta_rG^\circ(1) > 0; \Delta_rG^\circ(2) < 0$

Q5. JEE Main 2026 (21 January Shift 1)

Consider the following reactions.

$PbCl_2 + K_2CrO_4 \rightarrow A + 2KCl$

(Hot solution)

$A + NaOH \rightleftharpoons B + Na_2CrO_4$

$PbSO_4 + 4CH_3COONH_4 \rightarrow (NH_4)_2SO_4 + X$

In the above reactions, A, B and X are respectively.

(1) $Na_2[Pb(OH)_2], PbCrOO_4$ and $(NH_4)_2[Pb(CH_3COO)_4]$

(2) $Na_2[Pb(OH)_2], PbCrO_4$ and $[Pb(NH_3)_4]SO_4$

(3) $PbCrO_4, Na_2[Pb(OH)_4]$ and $(NH_4)_2[Pb(CH_3COO)_4]$

(4) $PbCrO_4, Na_2[Pb(OH)_4]$ and $[Pb(NH_3)_4]SO_4$

Q6. JEE Main 2026 (21 January Shift 1)

Given below are two statements :

Statement I : The number of pairs among $[SiO_2, CO_2]$, $[SnO, SnO_2]$, $[PbO, PbO_2]$ and $[GeO, GeO_2]$, which contain oxides that are both amphoteric is 2.

Statement II: BF_3 is an electron deficient molecule, can act as a Lewis acid, forms adduct with NH_3 and has a trigonal planar geometry.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false

(2) Both Statement I and Statement II are true

(3) Both Statement I and Statement II are false

(4) Statement I is false but Statement II is true

ANSWERS AND SOLUTIONS

1. (1) **2.** (4) **3.** (3) **4.** (3) **5.** (3) **6.** (2)

1. **(1)** Analyzing statements about Group 13 elements (B, Al, Ga, In, Tl):

A. Ionic radii of M^{3+} cations: Down a group, more electron shells are added, so ionic radii increase, not decrease. FALSE

B. Electronegativity: Generally decreases down the group, though Ga shows an anomaly with higher electronegativity than Al. The statement is approximately true but has exceptions.

C. First ionization enthalpy: B (800.6 kJ/mol) is highest due to smallest size and highest nuclear charge density. Al (577.5), Ga (579), In (558), Tl (589). TRUE

D. Trichlorides and triodides: BCl_3, $AlCl_3$, $GaCl_3$, etc. are all covalent compounds (electron-deficient). BI_3, AlI_3, GaI_3, etc. are also covalent (I^- is large and polarizable). TRUE

The clearly correct statements are C and D.

2. **(4)** The given reactions describe the Borax Bead Test.

When borax ($Na_2B_4O_7 \cdot 10H_2O$) is heated, it first loses water and then melts to form a transparent glassy bead consisting of sodium metaborate ($NaBO_2$) and boric anhydride (B_2O_3).

$Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2(X) + B_2O_3(Y)$

In the non-luminous (oxidizing) flame, B_2O_3 reacts with $CuSO_4$ to form copper(II) metaborate (Z), which is blue in color.

$CuSO_4 + B_2O_3 \xrightarrow{Non-Luminous} Cu(BO_2)_2(Z) + SO_3$

In $Cu(BO_2)_2$, the oxidation state of Cu is $+2$.

In the luminous (reducing) flame, the copper(II) metaborate is reduced by carbon to copper(I) metaborate (Q), which is colorless/red.

$2Cu(BO_2)_2(Z) + 2NaBO_2(X) + C \xrightarrow{Luminous} 2CuBO_2(Q) + Na_2B_4O_7 + CO$

In $CuBO_2$, the oxidation state of Cu is $+1$.

Therefore, the oxidation states of Cu in Z and Q are $+2$ and $+1$ respectively.

3. **(3)** For Statement I:

Among N, As, Sb, P, the electronegativity order is N > P > As > Sb. Thus X = N (most electronegative) and Y = Sb (least electronegative). N_2O_3 is acidic (forms HNO_2 with water) and Sb_2O_3 is amphoteric (reacts with both acids and bases).

Statement I is true.

For Statement II:

While BCl_3 is covalent and hydrolyzes in water to form $B(OH)_3$, the hydrolysis products are incorrect. In neutral aqueous medium, $B(OH)_3$ is formed. The tetrahedral $[B(OH)_4]^-$ forms only in basic conditions, and boron does

not form $[B(H_2O)_6]^{3+}$ (boron is not a typical transition metal).

Statement II is false.

4. **(3)** The stability of oxidation states in Group 14 is governed by the inert pair effect, which increases down the group.

For Lead (Pb), the $+2$ oxidation state is more stable than the $+4$ state ($Pb^{2+} > Pb^{4+}$).

In the reaction $PbO_2 + Pb \rightarrow 2PbO$, Pb^{4+} is being reduced to Pb^{2+} and Pb^0 is being oxidized to Pb^{2+}. Since the product contains the more stable oxidation state, the reaction is spontaneous, meaning $\Delta_r G^\circ(1) < 0$.

For Tin (Sn), the $+4$ oxidation state is more stable than the $+2$ state ($Sn^{4+} > Sn^{2+}$).

In the reaction $SnO_2 + Sn \rightarrow 2SnO$, the stable Sn^{4+} is being converted into the less stable Sn^{2+}. This process is non-spontaneous, meaning $\Delta_r G^\circ(2) > 0$.

Thus, the correct set is $\Delta_r G^\circ(1) < 0$ and $\Delta_r G^\circ(2) > 0$.

5. **(3)** Reaction 1: $PbCl_2 + K_2CrO_4 \rightarrow PbCrO_4 + 2KCl$

A = $PbCrO_4$ (lead chromate)

Reaction 2: $PbCrO_4 + 4NaOH \rightarrow Na_2[Pb(OH)_4] + Na_2CrO_4$

B = $Na_2[Pb(OH)_4]$ (sodium tetrahydroxoplumbate)

Reaction 3: $PbSO_4 + 4CH_3COONH_4 \rightarrow (NH_4)_2SO_4 + (NH_4)_2[Pb(CH_3COO)_4]$

X = $(NH_4)_2[Pb(CH_3COO)_4]$ (ammonium tetraacetatoplumbate)

6. **(2)** Statement I: Checking amphoteric nature of oxide pairs:

$[SiO_2, CO_2]$: Both acidic (non-metal oxides)

$[SnO, SnO_2]$: Both amphoteric

$[PbO, PbO_2]$: PbO is amphoteric but PbO_2 is predominantly acidic/oxidizing

$[GeO, GeO_2]$: Both amphoteric (Ge is metalloid)

2 pairs have both oxides amphoteric. Statement I is true.

Statement II: BF_3 has B with 6 valence electrons (electron deficient), acts as Lewis acid, forms $BF_3 \cdot NH_3$ adduct, has sp^2 hybridization giving trigonal planar geometry. Statement II is true.